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Potential energy surfaces near intersections

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The topographies of two potential energy surfaces are examined in the vicinity of their intersection. A brief account of the basic theory is given and the possible surface types are discussed explicitly. Two main patterns are found. One of these ("peaked") has the character of a tilted double cone in that the lower (upper) surface decreases (increases) in all directions from the intersection which is a point where an infinite number, in fact, all orthogonal trajectories emanate. The other pattern ("sloped") results when both surfaces are monotonically sloped and touch each other along the slope, with most orthogonal trajectories bypassing the intersection. When the latter pattern prevails, the intersection can lie on a steepest descent line which originates at a transition state and hence may qualify as a reaction path model. An intermediate pattern, involving a horizontal slope on both surfaces, is also possible. The topographical patterns also differ markedly with respect to the bunching of the steepest descent lines. In general, the latter tend to veer away from the intersection on the lower surface favoring bifurcations, but are funneled towards the intersection on the upper surface, making the vicinity of the intersection a region favoring radiationless transitions. The various cases are classified and illustrated through quantitative graphs of contours and orthogonal trajectories.

I. INTRODUCTION

Radiationless transitions among potential energy surfaces play essential roles in photophysical phenomena and photochemical reactions. They occur preferentially when potential energy surfaces (PES's) approach each other closely and, in particular, when they cross. Such crossings represent therefore an important aspect of potential energy surfaces. Although quantitative *ab initio* information about them is still limited, it has become apparent that intersections occur frequently. The results reported in previous papers furthermore show that they can lie close to transition states[1] as well as on steepest descent curves connecting transition states and minima,[2] i.e., on curves which are often viewed as models for reaction paths.

Fundamental theorems regarding PES intersections have been known since 1927[3] and have been discussed in further detail in recent years.[4-6] However, we are not aware of any quantitative presentation of the variety of shapes which PES's can assume around intersections nor of any discussion of their orthogonal trajectories. Since the difference between two PES's near an intersection is an elliptical cone, and since the two surfaces *together* form a double cone when the energy is plotted against the internal coordinate plane, it is customary to talk about "conical" intersections. However, this double cone may be arbitrarily tilted against the energy axis so that, in general, it is not too helpful for visualizing the energy contours on any one surface.

In the neighborhood of a point where two states cross, the first-order approximations to the two PES's depend essentially on three parameters, and their values strongly affect the topographical features of the surfaces. In the present note, we analyze and exhibit graphically contours as well as orthogonal trajectories for the various possible cases in this approximation. Since the values of these parameters can be determined readily in any specific case, the classification given provides adequate information for recognizing the character of a particular crossing.

II. THEORETICAL BACKGROUND

In the present section, some basic relationships are summarized, which are spread over the literature, and certain specific aspects are pursued in greater detail.

A. The crossing conditions

The intersection of two potential energy surfaces $E_1(\mathbf{q})$ and $E_2(\mathbf{q})$ implies that for certain values of the internal coordinates $\mathbf{q} = (q_1, q_2, \ldots, q_M)$ of the molecule, the two energies are equal. The locus of all points in the internal coordinate space where the identity $E_1(\mathbf{q}) = E_2(\mathbf{q})$ holds is a curvilinear coordinate *subspace* which we denote as the *intersection coordinate subspace* (ICS). If E_1 and E_2 were two *arbitrary* surfaces, then one would expect the condition $E_1 = E_2$ to determine an ICS of dimension $(M-1)$. However, E_1 and E_2 are not independent of each other, rather they are two solutions of the same eigenvalue problem of *one* Hamiltonian H. This circumstance leads to an additional condition, as was first recognized by von Neuman, Wigner, and Teller.[3]

Without loss of generality, the two intersecting adiabatic states ψ_1 and ψ_2 can be expressed as linear combinations of two *arbitrary*, e.g., diabatic orthogonal states ϕ_1 and ϕ_2 in the function space that is complementary to the space spanned by all *other* eigenstates ψ_n $(n \geqslant 3)$ of the relevant electronic Hamiltonian operator H whose energies $E_3, E_4, \ldots$ are assumed to be nondegenerate with E_1 and E_2:

$$\psi_1 = \phi_1 c_{11} + \phi_2 c_{21}, \quad \psi_2 = \phi_1 c_{12} + \phi_2 c_{22}. \tag{2.1}$$

The expansion coefficients c_{ji} and the corresponding ener-

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gies E_1, E_2 are determined by the eigenvalue problem of the 2×2 matrix $H_{ij} = \langle \phi_i | H | \phi_j \rangle$. We consider here the practically important case that all H_{ij} are real. The variation of E_j and c_{ij} with the nuclear motions is a result of the dependence of H_{ij} upon the M internal coordinates $\mathbf{q} = (q_1, q_2, \ldots, q_M)$ which describe the geometry of the nuclear framework: $H_{ij} = H_{ij}(\mathbf{q})$. The phase-consistent discussion of the solutions is made transparent by writing the matrix $\mathbf{H} = \{H_{ij}\}$ in the form

$$\mathbf{H} = \overline{H}\,\mathbf{I} + R\,\mathbf{U}, \tag{2.2}$$

with $\mathbf{I}$ denoting the identity matrix and $\mathbf{U}$ denoting the matrix

$$\mathbf{U} = \begin{pmatrix} \cos\alpha & \sin\alpha \\ \sin\alpha & -\cos\alpha \end{pmatrix}, \tag{2.3}$$

where the quantities $\overline{H}$, R, and α are defined by

$$\overline{H} = (H_{11} + H_{22})/2, \quad \Delta H = (H_{11} - H_{22})/2,$$
$$R = (\Delta H^2 + H_{12}^2)^{1/2}, \tag{2.4}$$
$$\cos\alpha = \Delta H/R, \quad \sin\alpha = H_{12}/R. \tag{2.5}$$

From Eqs. (2.2) and (2.3), the expectation value of an *arbitrary* linear combination $\psi = (\phi_1 \cos\gamma + \phi_2 \sin\gamma)$ is found to be

$$\langle \psi | H | \psi \rangle = \overline{H} + R\cos(2\gamma - \alpha). \tag{2.6}$$

The manifest extrema of this function with respect to γ yield the two eigenstates

$$\psi_1 = \phi_1 \cos\alpha/2 + \phi_2 \sin\alpha/2 \tag{2.7}$$

with

$$E_1 = \overline{H} + R, \tag{2.8}$$

and

$$\psi_2 = \phi_1 \cos(\alpha + \pi)/2 + \phi_2 \sin(\alpha + \pi)/2$$
$$= -\phi_1 \sin\alpha/2 + \phi_2 \cos\alpha/2 \tag{2.9}$$

with

$$E_2 = \overline{H} - R. \tag{2.10}$$

According to Eqs. (2.8) and (2.10), an intersection occurs whenever R vanishes, which according to Eq. (2.4), requires that the *two* quantities ΔH and H_{12} *both* vanish. The intersection coordinate subspace (ICS) is therefore defined by the two crossing conditions

$$\Delta H(\mathbf{q}) = 0, \quad \text{i.e., } H_{11}(\mathbf{q}) = H_{22}(\mathbf{q}), \tag{2.11}$$
$$H_{12}(\mathbf{q}) = 0. \tag{2.12}$$

B. The intersection space (ICS)

1. Dimension and symmetry

It is possible that the equation system (2.11) and (2.12) has no real solutions, in which case the two states do not cross. However, *if* a real solution exists for one particular point $\mathbf{q}^0$, *then* it is to be expected (i.e., the opposite is unlikely) that this point is part of a contiguous $(M-2)$ dimensional region in which this remains true. This inference is based on the premise that the matrix elements H_{ij} are piecewise continuous functions in all coordinates. Thus, *if* there exists an ICS, *then* it is expected to be of dimension $(M-2)$.

This conclusion has to be modified in those cases where, for some reason, the off-diagonal element H_{12} vanishes *identically* for all values of the nuclear coordinates $q_1, q_2, \ldots, q_M$. In such a case, ϕ_1 and ϕ_2 are clearly the eigenfunctions ψ_1, ψ_2 for all values of $\mathbf{q}$ and Eq. (2.11) is the only remaining condition for a crossing. Thus, the ICS, if it exists, is expected to be of dimension $(M-1)$. The most common reason for this to happen is that the two states ψ_1, ψ_2 belong to different irreducible representations of the symmetry group of the molecule. In such a case, the space spanned by ϕ_1 and ϕ_2 spans a reducible representation and the reduction, by purely group theoretical means, yields two states belonging to different irreps between which the off-diagonal element H_{12} vanishes identically for all $(q_1, q_2, \ldots, q_M)$. Hence, they are in fact the eigenstates ψ_1, ψ_2 and Eq. (2.11) remains as only a condition for crossing so that the ICS is of dimension $(M-1)$.

Often a situation is found which combines aspects of the two cases just discussed. It occurs in molecules where a certain restricted class of nuclear arrangements belongs to a higher symmetry group than other geometries. For example, a molecule XY_2 belongs to the symmetry group C_s when all internuclear distances are different, but it belongs to the higher group C_{2v} when the two (XY) bond lengths are the same. In such molecules, there usually exist states which belong to the same irreducible representation in the lower symmetry (e.g., A' in C_s), but to different irreps of the higher symmetry (e.g., A_1 and B_1 in C_{2v}) for the restricted class of geometries. There are many examples for two such states to cross for a geometry that has the higher symmetry. In the space of higher symmetry, the upper state belongs to one irrep (e.g., A_1) on one side of the intersection and to the other irrep (e.g., B_1) on the other side of the intersection. The reverse is true for the lower state. If the restricted coordinate space of higher symmetry is of dimension R, then the M coordinates can be chosen in such a manner that all variations $(q_1, q_2, \ldots, q_R)$ maintain the higher symmetry and $(q_{R+1}, q_{R+2}, \ldots, q_M)$ break that symmetry unless they all vanish. It can then be concluded that the intersection space of the two states is of dimension $(M-2)$ in the *total* space and of dimension $(R-1)$ in the *restricted* space of higher symmetry. The intersection space is therefore of dimension $(R-1)$ in the coordinates $(q_1, \ldots, q_R)$, and has $[(M-2)-(R-1)] = (M-R-1)$ additional degrees of freedom when the coordinates $(q_{R+1}, \ldots, q_M)$ are allowed to vary as well. If it so happens that $R = M-1$, then it is thus possible that the intersection space lies entirely in the restricted coordinate space of higher symmetry.

The discussed case of two states belonging to the same irrep in the larger space of low symmetry and to different irreps in the restricted space of high symmetry is not the only possibility for the intersection of two PES's in a space of higher symmetry. It can also happen that two PES's which intersect in the space of higher symmetry belong to the same irrep in the space of higher symmetry as well as in the space of lower symmetry. An example of such a crossing has recently been found through *ab initio* calculations for the lowest two $^1A_1$ states of ozone.[1] In terms of the coordinates cho-

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sen at the end of the last paragraph, the ICS is, in such cases, of dimension $(M-2)$ in the total space, of dimension $(R-2)$ in the coordinates $(q_1,q_2,\ldots,q_R)$ and has $(M-2)-(R-2)=(M-R)$ additional degrees of freedom when the coordinates $(q_{R+1},q_{R+2},\ldots,q_M)$ are added to the coordinates $(q_1,q_2,\ldots,q_R)$. That is, the intersection space corresponds to a subspace of dimension $(R-2)$ in the restricted coordinate space of $q_1,\ldots,q_R$, complemented by the full variation in the coordinates $q_{R+1},\ldots,q_M$.

Still another situation is found in the cases of the Jahn–Teller and the Renner–Teller effects. Here two states, which belong to *different* irreps in the larger space of low symmetry, become two partner functions spanning *one* two-dimensional irrep in a restricted space where a higher, non-Abelian symmetry group applies. For the Renner–Teller effect, this group is $C_{\infty v}$; for the Jahn–Teller effect, it is another point group.

Diatomic and triatomic molecules are special in that, for them, the lowest possible symmetry is not C_1 as it is for all other molecules. In triatomic molecules, it is C_s, so that, for *any* molecular geometry, there exist two kinds of states, those belonging to the irrep A' and those belonging to A''. Since there are three internal coordinates, the intersection "seam" between two surfaces is a line (dimension 3-2) if they belong to the same irrep and a two-dimensional sheet (dimension 3-1) if they belong to different irreps. Furthermore, within the full three-dimensional coordinate space, there always exists a two-dimensional subspace, corresponding to the linear conformations of the molecule, where it belongs to $C_{\infty h}$ symmetry. Since most of its irreps are two-dimensional, there exists then the possibility for a Renner–Teller effect. When the two atoms are of the same element, then there also exists a two-dimensional coordinate subspace where the molecule has C_{2v} symmetry. One then has three possible cases: (i) both states belong to the *same* irrep in C_s *and* C_{2v} (in this case, the ICS seam is a line in the full C_s space and penetrates the C_{2v} coordinate plane in a point); (ii) both states belong to *different* irreps in C_s *and* C_{2v} (in this case, the ICS is a two-dimensional sheet in C_s and penetrates the C_{2v} coordinate plane in a line); (iii) the two states belong to the *same* irrep in C_s, but to *different* irreps in C_{2v} (in this case, the ICS seam is a line in C_s *and* in C_{2v}; it is therefore possible that it is a line which lies *entirely* in the C_{2v}-preserving coordinate space without extending in the full C_s space). If all three atoms are of the same element, then there exists a one-dimensional coordinate subspace of equilateral triangles with D_{3h} symmetry. Since this group has two-dimensional irreps, one has here the possibility of a Jahn–Teller effect for two states belonging to different symmetry in C_{2v}.

Diatomic molecules are special in that they have only one internal coordinate, i.e., the internuclear distance, and in that they belong to the symmetry groups $C_{\infty v}$ or $D_{\infty h}$. The states may therefore belong to many different irreps. Two states belonging to different irreps intersect a point (dimension $1-1=0$). However, for two states of the same symmetry, the *two* conditions (2.11) and (2.12) cannot, in general, be satisfied by adjusting the one available variable R and hence they cannot be expected to intersect. This "non-crossing rule" is, however, strictly limited to diatomic molecules. Attempts to establish a wider validity for it have failed.[5]

2. Intersection space and reaction paths

At a transition state, orthogonal trajectories of PES contours emanate in the directions of the M normal modes. One of them, the intrinsic reaction path, is a steepest descent line. The $(M-1)$ others are steepest ascent lines and define the barrier ridge. If the transition state lies in an R-dimensional coordinate subspace of higher symmetry, then its normal modes either lie entirely in the space of higher symmetry or are perpendicular to it. It can then be shown that the orthogonal trajectories starting out *in* the space of higher symmetry maintain that symmetry until they come to a point where the gradient vanishes or is discontinuous. For example, if the transition state of a triatomic molecule lies in the C_{2v} preserving subspace, then two of the orthogonal trajectories emanating from the transition state stay in the C_{2v} subspace and the third one starts out along the C_{2v} symmetry breaking coordinate.

If now, in the subspace of higher symmetry, this PES also has an intersection with another PES belonging to a *different* irreducible representation, then the intersection space is of dimension $(R-1)$ within the higher symmetry space. It is therefore not unlikely that any one of the R one-dimensional orthogonal trajectories originating at the transition state in the high-symmetry subspace (be it the reaction path or one of the ridge lines) crosses the ICS. Indeed, we have shown elsewhere[2] that the reaction path of the cyclopropylidene ring opening goes through an intersection after having passed the transition state under the described conditions.

If the PES intersects another surface belonging to the *same* irreducible representation, then the ICS is of dimension $(R-2)$ within the high symmetry subspace and one would think that it is unlikely to be crossed by a specific one-dimensional orthogonal trajectory in that subspace. For example, in the C_{2v} preserving subspace of a triatomic molecule, such an intersection is a *point* in a two-dimensional plane and it would seem improbable that an orthogonal trajectory in that plane would go through exactly that point. Nonetheless, we have found[1] that in the ozone molecule, such an intersection point does in fact lie on the orthogonal trajectory emanating from a transition state along a barrier ridge. In addition, this intersection point lies within 0.04 Å and 1 kcal/mol of a transition state on one PES and a minimum on the other PES. These "coincidences" are clearly consequences of an intrinsic relation between the lowest two eigenvalues of this particular Hamiltonian, so that purely probabilistic reasoning is inappropriate.

3. Intersection adapted coordinates

The discussion of an intersection is simplified by choosing, in a ceratin domain around the intersection, curvilinear coordinates which are particularly suited to the problem. Such intersection adapted coordinates $p_1,p_2,p_3,\ldots,p_M$ can be defined as follows:

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$$p_1 = H_{12}(\mathbf{q}), \tag{2.13a}$$

$$p_2 = \Delta H(\mathbf{q}) = [H_{11}(\mathbf{q}) - H_{22}(\mathbf{q})]/2, \tag{2.13b}$$

$$p_j = p_j(\mathbf{q}), \quad j = 3,4,\ldots,M, \tag{2.13c}$$

where p_j for $j \geqslant 3$ are chosen orthogonal to p_1, p_2 in the sense that

$$\sum_{t=1}^{M} (\partial p_j/\partial q_t)(\partial p_n/\partial q_t) = 0 \tag{2.14}$$

for $n = 1,2$ and $j = 3,4,\ldots,M$. In the domain where these definitions are valid, the two energy surfaces are given by

$$E_{1,2} = \overline{H}(\mathbf{p}) \pm (p_1^2 + p_2^2)^{1/2} \tag{2.15}$$

and the intersection space is then given by the conditions $p_1 = p_2 = 0$. It is therefore spanned by the coordinates $p_3, p_4, \ldots, p_M$.

4. Tangent intersection space

It is useful to examine the ICS within a small neighborhood of a point $\mathbf{q}^0 = (q_1^0, q_2^0, \ldots, q_M^0)$. In such a domain, the ICS is approximated by the $(M-2)$ dimensional *linear* space which is tangential to the ICS at $\mathbf{q}^0$, the *tangent* ICS. It is obtained by expanding the matrix elements $H_{ij}(\mathbf{q})$ to first order in terms of the displacements $\mathbf{x}$ from the point $\mathbf{q}^0$, i.e.,

$$\mathbf{x} = \mathbf{q} - \mathbf{q}^0 = (x_1, x_2, \ldots, x_M). \tag{2.16}$$

Writing

$$\overline{H}(\mathbf{q}) = \overline{H}^0 + \sum_{t=1}^{M} U_t x_t,$$

$$\overline{H}^0 = \overline{H}(\mathbf{q}^0), \quad U_t = (\partial \overline{H}/\partial q_t)^0, \tag{2.17a}$$

$$\Delta H = 0 + \sum_{t=1}^{M} V_t x_t, \quad V_t = [\partial(\Delta H)/\partial q_t]^0, \tag{2.17b}$$

$$H_{12} = 0 + \sum_{t=1}^{M} W_t x_t, \quad W_t = (\partial H_{12}/\partial q_t)^0, \tag{2.17c}$$

we can express the two energy surfaces near $\mathbf{q}_0$ as

$$E_{1,2} = \sum_t U_t x_t \pm \left[\left(\sum_t V_t x_t\right)^2 + \left(\sum_t W_t x_t\right)^2\right]^{1/2}. \tag{2.18}$$

The tangent ICS is then determined by

$$\sum_t V_t x_t = \sum_t W_t x_t = 0, \tag{2.19}$$

i.e., it is the plane orthogonal to the gradients of H_{12} and ΔH.

C. The branching space

1. Energies

While the degeneracy $E_1 = E_2$ is *preserved* in the $(M-2)$ dimensional ICS spanned by the coordinates $p_3, \ldots, p_M$, it is *lifted* in the two-dimensional *branching space* spanned by the coordinates p_1, p_2. On this sheet, defined by $p_3 = p_4 = \cdots = p_M = 0$, the energies are given by

$$E_{1,2} = \overline{H}(p_1, p_2) \pm (p_1^2 + p_2^2)^{1/2}. \tag{2.20}$$

Near a point $\mathbf{q}_0$ on the ICS, this branching sheet is approximated by the *tangent branching plane* spanned by the gradients of H_{12} and ΔH at $\mathbf{q}_0$, i.e., by the coordinates

$$y_1 = \sum_t V_t x_t \approx p_1, \quad y_2 = \sum_t W_t x_t \approx p_2. \tag{2.21}$$

In this tangent branching space, the energies become

$$E_{1,2} \approx A_1 y_1 + A_2 y_2 \pm (y_1^2 + y_2^2)^{1/2}. \tag{2.22}$$

On any given, fixed straight line through the intersection in the tangent branching space, say $y_1 = B_1 t$, $y_2 = B_2 t$, one finds

$$E_1 = (A_1 B_1 + A_2 B_2)t + (B_1^2 + B_2^2)^{1/2}|t|,$$

for the upper state,

$$E_2 = (A_1 B_1 + A_2 B_2)t - (B_1^2 + B_2^2)^{1/2}|t|,$$

for the lower state,

so that one has the identities

$$(\partial E_1/\partial t)^0_- = (\partial E_2/\partial t)^0_+, \tag{2.23a}$$

$$(\partial E_2/\partial t)^0_- = (\partial E_2/\partial t)^0_+. \tag{2.23b}$$

They are illustrated by Fig. 1, which also explains the somewhat confusing terms "intersection" and "crossing." They are based on the picture that on any line through the intersection, $(E_1)_+$ and $(E_2)_-$ form a continuously varying state with a continuous slope [see also the remarks after Eq. (2.25) regarding the continuity of the wave functions] as do $(E_1)_-$ and $(E_2)_+$. In fact, $(E_1)_+$ and $(E_1)_-$ are of course connected by going around the crossing in an additional dimension as are $(E_2)_+$ and $(E_2)_-$. Figure 1 also shows that the two surfaces can connect in three possible ways, namely, according to a "peaked" pattern, a "sloped" pattern, and an intermediate, "semilevel" pattern, a distinction which will turn out to be useful for the classification of intersections to be developed in Sec. II. The stated conclusions remain valid if the straight line also has components with respect to the additional coordinates $p_3, p_4, \ldots, p_M$.

2. Wave functions

The quantities R and α which were introduced in Eqs. (2.4) and (2.5) have a simple meaning in the branching space. From the definitions (2.13), it follows that

$$R = (p_1^2 + p_2^2)^{1/2}, \quad \cos\alpha = p_1/R, \quad \sin\alpha = p_2/R \tag{2.24}$$

and it is apparent that *(R,α) are the polar coordinates corresponding to the intersection-adapted coordinates (p_1, p_2) if the latter are plotted as a Cartesian axis system.* On the other hand, the states ψ_1, ψ_2 can be considered as unit vectors in the function space spanned by the orthogonal functions ϕ_1, ϕ_2. According to Eqs. (2.7) and (2.9), *the polar angles in this function space are $(\alpha/2)$ for ψ_1 and $(\alpha + \pi)/2$ for ψ_2.* Con-

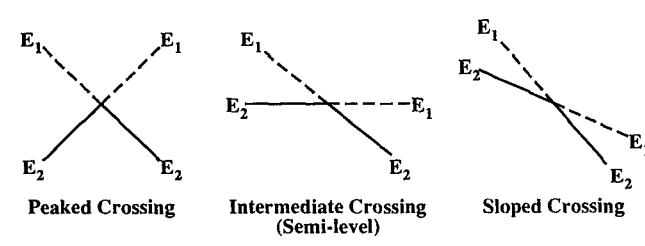

FIG. 1. The "exchange of slopes" between two potential energy surfaces and the three basic patterns at an intersection.

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sequently, if one moves along a continuous sequence of coordinates (p_1,p_2) describing a loop around the intersection point $\mathbf{q}^0$ in the coordinate branching space, then the two corresponding eigenstates rotate continuously around the origin in the (ϕ_1,ϕ_2) function space, but "at half the speed." This is illustrated in Fig. 2. Since ϕ_1 and ϕ_2 can be assumed to be univalued everywhere, the following two inferences can therefore be drawn:

First, if one travels halfway around $\mathbf{q}^0$, corresponding to an increase in α by π, then one would have

$$\psi_1(\alpha+\pi)=\psi_2(\alpha),\quad \psi_2(\alpha+\pi)=-\psi_1(\alpha) \quad (2.25)$$

if the functions ϕ_1 and ϕ_2 were the same. This becomes in fact the case *at* the intersection. Thus, Eq. (2.25) holds when ψ_1 is the limiting wave function immediately "before" and ψ_2 is the limiting wave function immediately "after" passing through $\mathbf{q}^0$ on a straight line ("before" corresponding to α and "after" corresponding $\alpha+\pi$). At the intersection point, the wave function of the lower PES connects thus with that of the higher PES and vice versa, except for possible changes in sign. Consequently, in passing through the intersection along any fixed direction, the wave functions of both, the lower as well as the higher PES, change their character abruptly because, in fact, they *exchange characters* at $\mathbf{q}$. (*At* the intersection, *any* linear combination of ψ_1 and ψ_2 is of course an eigenfunction.) This behavior of the wave function is related to the behavior of the gradients of E_1 and E_2 expressed by Eq. (2.23) and depicted in Fig. 1 and is the limiting case of the behavior found for wave functions near avoided crossings.

The sudden change in the character of the wave function of the lower PES and that of the higher PES is very obvious for the intersection of two surfaces of different symmetries because, if the PES belonging to irrep A lies above that for irrep B "before" the intersection, then the reverse is true "after" the intersection in such a case. Consequently, both the upper as well as the lower PES change from one irrep to the other when one moves through the intersection.

There exists some inconsistency in the use of the concept of a "state" in such situations. In general and in the case of two PES's of the same symmetry, one state is usually defined as comprising all wave functions with the lower energy, the other state is defined as comprising all wave functions with the higher energy. However, when the two states belong to different irreducible representations of a group, then it is customary to consider all energies corresponding to wave functions *belonging to the same irreducible representation* as one state and one PES. The second convention contradicts the first one. The contradiction leads to a conceptual conflict in cases, such as discussed above, when the two surfaces belong to the same irrep in the lower symmetry group, which is valid in the entire coordinate space, but to different irreps in a restricted coordinate subspace where a higher symmetry holds. In such cases, it seems appropriate to stick with the first convention and to admit that, in the high symmetry subspace, each state belongs to one irrep in one domain and to another irrep in another domain.

The second inference of Fig. 2 is the theorem of Herzberg and Longuet-Higgins[4] which states: if one follows the

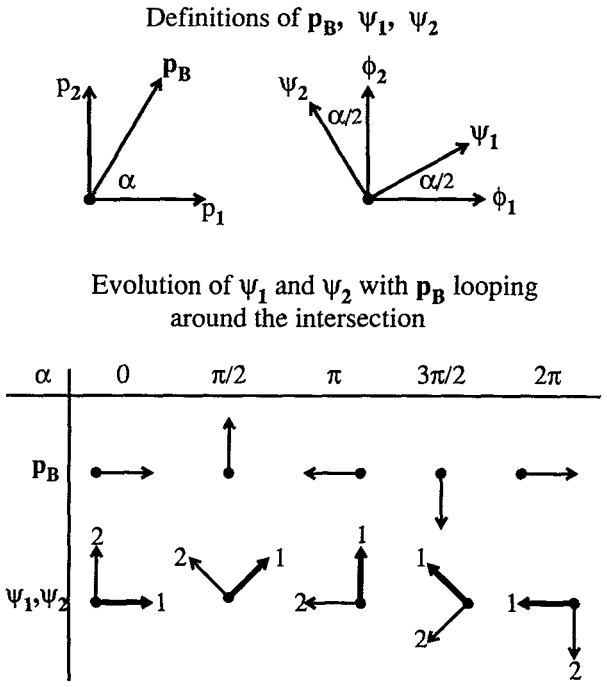

FIG. 2. The evolution of the two eigenfunctions ψ_1, ψ_2 in function space along an internal coordinate loop around the intersection. $\mathbf{P}_B$ = projection of the intersection-adapted internal coordinate vector in the branching plane p_1, p_2.

continuous deformations of the wave functions ψ_1, ψ_2 along a *closed* path looping around the intersection point $\mathbf{q}^0$, then both wave functions return to their original form multiplied by (-1), i.e.,

$$\psi_1(\alpha+2\pi)=-\psi_1(\alpha),\quad \psi_2(\alpha+2\pi)=-\psi_2(\alpha). \quad (2.26)$$

This theorem permits the determination of the existence of an intersection in a certain domain without having to find its location or when the equality of the energies of the two states has only been approximately established.[1]

It is evident that the two discussed inferences remain valid: (i) if one follows the straight line through $\mathbf{q}_0$ or the continuous loop around $\mathbf{q}_0$, respectively, in terms of any other coordinates which are related to p_1,p_2 by a one-to-one topological mapping, and (ii) if one extends the path followed by allowing the variables $p_3,p_4,\ldots,p_M$ to become different from zero and letting them vary at the same time, as long as the variables p_1,p_2 remain sufficiently well defined along the chosen path. It is also apparent that the validity of the phase theorem is limited to the domain in which the functions $H_{12}(q)$ and $\Delta H(q)$ are sufficiently regular and where a one-to-one correspondence exists between the intersection adapted coordinates $\mathbf{p}$ and the original coordinates $\mathbf{q}$.

III. TOPOGRAPHY OF THE LOWER SURFACE IN THE TANGENT BRANCHING SPACE

If the branching space is of dimension 1, then the three patterns of Fig. 1. cover all possibilities and there is not much to discuss. In the present section, we consider PES's in a two-dimensional branching space, which is the general case.

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A. Expression for a PES in terms of mass-weighted Cartesian coordinates

While the difference between the two potential energy surfaces in the tangent branching space

$$E_1 - E_2 = 2(y_1^2 + y_2^2)^{1/2} \tag{3.1}$$

is a cone, the individual PES's of Eq. (2.22) are superpositions of a plane and a cone. In the present section, we shall examine the basic topography of such functions. We shall also examine their steepest descent lines which furnish additional information and are often considered as model curves for reaction paths. Since orthogonal trajectory patterns depend markedly on coordinate scaling, it is relevant that reaction paths are generally believed to be most appropriately modeled by steepest descent lines in the space of mass-weighted Cartesian coordinates or coordinates obtained from them by orthogonal transformations. The coordinates which led to the simple Eq. (2.22) were, however, obtained by a very specific scaling which, in general, is not orthonormal. In terms of displacements $x_1, x_2, \ldots, x_M$ of mass-weighted Cartesian coordinates, the expressions for the two PES's have therefore the more general form

$$E_{1,2} - E^0 = B_1 x_1 + B_2 x_2 \pm (C_1 x_1^2 + C_2 x_2^2 + 2C_{12} x_1 x_2)^{1/2}, \tag{3.2}$$

where the quadratic under the radical sign is positive definite. It can be brought into diagonal form by an orthogonal transformation and, in terms of these new coordinates, one has

$$E_{1,2} - E^0 = b_1 x + b_2 y \pm (c_1^2 x^2 + c_2^2 y^2)^{1/2}. \tag{3.3}$$

Without loss of generality, we can divide through by c_1, which can be assumed positive, and obtain

$$E_2 - E^0 = c_1 f(x,y), \quad \text{for the lower PES,} \tag{3.4a}$$

$$E_1 - E^0 = -c_1 f(-x,-y), \quad \text{for the upper PES,} \tag{3.4b}$$

where

$$f(x,y) = -ax - by - (x^2 + c^2 y^2)^{1/2}. \tag{3.5}$$

Since both the upper and the lower PES's can be obtained from the function $f(x,y)$, we shall discuss this function, i.e., the lower surface, in this section in detail. It is apparent that the surfaces obtained for the parameters a and $(-a)$ are each others' mirror images with respect to the y axis and that the surfaces obtained for b and $(-b)$ are each other's mirror images with respect to the x axis. We therefore limit the parameters a and b to positive values. This has the effect that in those cases where f is monotonic, it *decreases* with increasing values of x and y.

The energy of the lower surface, plotted as a function of x and y, is an elliptical cone tipped over. The contour curves of constant energy [$f(x,y) = \text{constant}$] are intersections of planes with this cone, i.e., they are conic sections. Depending on the ellipticity and how the cone is tipped, they may be ellipses, parabolas, hyperbolas, or limiting cases such as circles or straight lines.

B. The case of a circular difference cone

1. General considerations

We begin by examining the special case of a circular difference cone because it is simple and yet exhibits important characteristic features. According to Eq. (3.5), a circular difference cone results when the parameter c is unity. Since the radicand $(x^2 + y^2)$ is invariant under orthogonal transformations, it is apparent that the expression for $f(x,y)$ can be brought into the form

$$f(x,y) = -ax - (x^2 + y^2)^{1/2} \tag{3.6}$$

while still maintaining orthogonally transformed mass-weighted Cartesian coordinates. Thus, we can assume $b = 0$ without loss of generality. [When *arbitrary* scaling is admitted, as in Eq. (2.22), *all* surfaces reduce to this form.]

Contour plots of the function (3.6) are exhibited in Fig. 3. Nine panels are shown corresponding to nine values of the parameter a. In all panels, both coordinates x and y go from -2.5 to $+2.5$. The origin, indicated by a bold-faced dot at the center, is the intersection point. Solid lines depict the contours $f = 0.3n$ (n = integer). Bold-faced solid lines are contours where $f = 0$. The functions $f(x,y)$ are positive in the shaded areas and negative in the unshaded areas. Dashed lines depict orthogonal trajectories. Bold-faced dashed lines indicate orthogonal trajectories that are straight lines. They separate orthogonal trajectories of opposite curvatures from each other.

2. Sloped patterns

It is apparent from Eq. (3.6) that there are always regions where f is negative. If there also exist areas where f is positive, then there must be contours for which $f(x,y) = 0$. According to Eq. (3.6), they are given by

$$ax = -(x^2 + y^2)^{1/2}. \tag{3.7a}$$

When $a > 1$, then Eq. (3.7a) yields the two straight lines

$$y = \pm (a^2 - 1)^{1/2} x. \tag{3.7b}$$

Since a is assumed to be positive, Eq. (3.7a) also shows, however, that x must be negative. Equation (3.7b) defines therefore two straight *half-lines* which meet at the origin, as indeed shown in Fig. 3 on the four panels for $a > 1$. The two bold-faced half-lines corresponding to $f = 0$ form *one* contour separating the shaded domain at the left where $f > 0$ from the domain at the right where $f < 0$.

These panels are labeled by the symbol S indicating that all of these surfaces are "sloped" in the sense that they descend monotonically from left to right, i.e., with increasing values of x. The slope of this descent is much steeper in the region of negative f than in the region of positive f, i.e., the surfaces decrease gently coming from the left and rapidly to the right of $f = 0$. From Eq. (3.6), one obtains for the orthogonal trajectories the differential equation

$$dy/dx = y/[x + a(x^2 + y^2)^{1/2}] \tag{3.8}$$

whose numerical integration yields the dashed steepest descent lines in Fig. 3. All of them originate in the region to the left where f is positive and, for large negative x values, start out parallel to the x axis. As x approaches zero, the orthogonal trajectories turn away from the x axis so as to cut through

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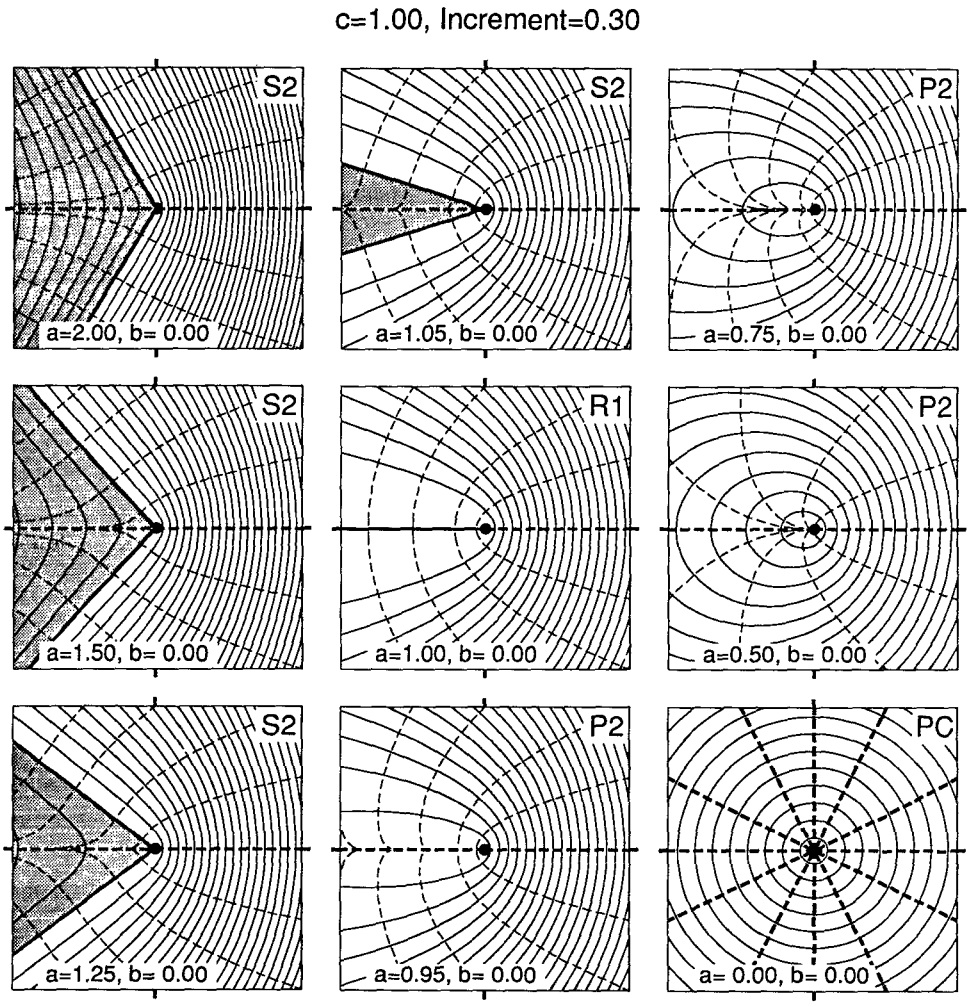

FIG. 3. Potential energy surface patterns near an intersection for the case $c = 1$, $b = 0$. Intersection = heavy dot at center. Ranges of abscissa and ordinate: $-2.5 < x < 2.5$, $-2.5 < y < 2.5$; solid lines: contour lines; bold-faced solid lines: $E - E_0 = 0$; shaded area: $E - E_0 > 0$; contour increment = 0.3; dashed lines: orthogonal trajectories; bold-faced dashed lines: orthogonal trajectories which are straight lines.

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the bold-faced straight line contours $f = 0$ at right angles. The closer to the negative x axis the steepest descent lines start out, the more abruptly they turn away from the x axis at the appropriate points.

The x axis itself is a steepest descent line which rides downhill on a ridge from left to right straight through the intersection point, where its slope abruptly increases in absolute value. It is readily confirmed that $y = 0$ is in fact the only straight line which satisfies the differential equation (3.8). Since the slope changes discontinuously at the origin, we consider it as two half-lines, one ending and the other beginning at the origin. The existence of two straight-line orthogonal trajectories is indicated by the number 2 in the symbol $S2$ labeling these panels. By virtue of Eq. (3.4), it is apparent that the steepest descent lines of E_1 and E_2 going through the intersection form the sloped pattern of Fig. 1.

We have discussed the pattern for $c = 1$, $a > 1$ in some detail since, in a previous investigation,[1] we found that a reaction path descending from a transition state was modeled by a steepest descent line passing through an intersection in exactly the same manner as the x axis does in the "S2" panels of Fig. 3. Immediately after the transition state, this steepest descent line followed the floor of a valley. Then, at a *valley–ridge inflection point*,[7] it turned into the crest of a ridge which then passed through the intersection as illustrated in Fig. 3. Upon approaching the intersection point on such a reaction path, the system will become subject to random dynamic instabilities when it arrives on the ridge and will fall off it. The orthogonal trajectory pattern exhibited in Fig. 3 implies that the system once off the ridge will be subject to forces which will push it more and more away from the ridge and more and more strongly so. It is thus apparent that in the region around an intersection point with the discussed topography, *the PES has a strong bifurcating effect on the reaction path*, and particularly so when the parameter a is close to 1.

3. Peaked patterns

When $a < 1$, then Eq. (3.7) has only one solution, namely, $x = y = 0$. Thus, the function $f(x,y)$ vanishes only at the origin and descends in all directions towards negative values. This peaked pattern is therefore labeled by the symbol P. The positive as well as the negative x axis both are again the only straight-line orthogonal trajectories and the panels are therefore labeled as $P2$. All other steepest descent lines start out at the origin in the direction of the *negative* x axis and then turn around towards increasing x values. The contours

are ellipses. The steepest descent lines on E_1 and E_2 form the peaked pattern of Fig. 1.

When $a = 0$, these contours become circles and *all* orthogonal trajectories become straight lines. The function $f(x,y)$ is a circular cone and the panel is therefore labeled as PC in Fig. 3.

4. Intermediate patterns

For $a > 1$, the inclination of the bold-faced contours $f = 0$ with respect to the x axis becomes progressively smaller as the parameter a approaches 1. Concommitantly, the region of positive f values becomes progressively narrower and the steepest descent lines start out as a progressively narrower bundle to the left, which upon approaching the origin, suddenly splits by veering to one side on the other. For $a = 1$, we have the limiting case between the patterns P and S, and we denote this intermediate pattern as R. In this case, the two bold-faced straight-line contours $f = 0$ of the S pattern have collapsed into the negative x axis and the region with $f > 0$ has been squeezed out. The function f descends downhill from all points on the negative x axis, which is now the contour $f = 0$. Hence, only the positive x axis is a straight-line orthogonal trajectory and the panel symbol is therefore R 1. All steepest descent lines, except the $+x$ axis, start out perpendicular to the $-x$ axis and are mirror images (with respect to the y axis) of certain contour lines of $f(x,y)$. The steepest descent lines on E_1 and E_2 form the intermediate pattern of Fig. 1.

C. The case of an elliptic difference cone with principal axes lined up along the linear term

1. General considerations

For c different from unity, the difference cone $(E_1 - E_2)$ is elliptical and, in general, its principal axes are unrelated to the vector (a,b) which determines the linear term. However, when $b = 0$, then

$$f(x,y) = -ax - (x^2 + c^2y^2)^{1/2} \tag{3.9}$$

and the vector $(a,b) = (a,0)$ is lined up with one of the axes of the difference cone. It is unnecessary to consider the case $(a = 0, b \neq 0)$ as well, if we consider $c > 1$ *and* $c < 1$ for the case $(a \neq 0, b = 0)$. This is because rotation by 90° will map each pattern $(a = 0, b \neq 0, c > 1)$ into a pattern $(a \neq 0, b = 0, c < 1)$ and each pattern $(a = 0, b \neq 0, c < 1)$ into a pattern $(a \neq 0, b = 0, c > 1)$.

As in the case of a circular difference cone, we have regions where $f > 0$ only if there exist contours $f = 0$. From Eq. (3.9), it follows that they are given by

$$y = \pm [(a^2 - 1)^{1/2}/c]x, \quad x < 0 \tag{3.10}$$

provided that $a \geqslant 1$, as before. A region $f > 0$ exists when $a > 1$ and it lies between the two straight half-lines defined by Eq. (3.10), as illustrated by the bold-faced solid lines in Fig. 4. For $a = 1$, the negative x axis with $f = 0$ is the crest of the surface. For $a < 1$, the origin is the only point where $f = 0$. Accordingly, the three cases $a > 1$, $a = 1$, and $a < 1$ are again characterized by the symbols P, R, and S introduced in the last section.

The orthogonal trajectories are now given by the differential equation

$$dy/dx = c^2y/[x + a(x^2 + c^2y^2)^{1/2}]\,. \tag{3.11}$$

As before, the dashed lines in Fig. 4 represent some of them, obtained by numerical integration, and the bold-faced dashed lines are straight-line orthogonal trajectories. The equations for the latter are obtained by substituting $y = kx$ into Eq. (3.11):

$$k = c^2kx/[x + a(x^2 + c^2k^2x^2)^{1/2}],$$

$$k\,[1 + \mathrm{sign}(x)a(1 + c^2k^2)^{1/2} - c^2] = 0,$$

which factors into the two equations

$$k = 0, \tag{3.12}$$

$$(1 + c^2k^2)^{1/2} = [(c^2 - 1)/a]\mathrm{sign}(x). \tag{3.13}$$

Thus, we again have the orthogonal trajectories $y = 0$ for $x > 0$ and for $x < 0$ (except in the case $a = 1$, where $x > 0$ is the only one). However, in addition, we find now from Eq. (3.13) the two straight-line orthogonal trajectories given by

$$k = \pm [(c^2 - 1)^2 - a^2]^{1/2}/ac, \tag{3.14a}$$

$$\mathrm{sign}(x) = \mathrm{sign}(c^2 - 1) \tag{3.14b}$$

provided that $|c^2 - 1| > a$.

2. The x axis is the shorter principal axis of the difference cone (c < 1)

The case $c < 1$ is illustrated in Fig. 4 for $c = 0.5$. As discussed after Eq. (3.10), regions with $f > 0$ exist for $a > 1$ (panels S), but not for $a < 1$ (panels P). Panel R is the limiting case.

All panels S and P exhibit the two straight-line orthogonal trajectories $(y = 0, x > 0)$ and $(y = 0, x < 0)$. However, it is readily verified that for the panel R, corresponding to $a = 1$, only $(y = 0, x > 0)$ is an orthogonal trajectory.

In addition, there exist, however, the additional straight-line orthogonal trajectories corresponding to Eq. (3.14), as long as $a < |c^2 - 1| = 1 - c^2 = 0.75$. In agreement with Eq. (3.14b), they are half-lines with $\mathrm{sign}(x) = \mathrm{sign}(c^2 - 1) < 0$.

Thus, there are four straight-line orthogonal trajectories for $a < 1 - c^2 = 0.75$ and two straight-line orthogonal trajectories for $a \geqslant 0.75$, except only one such orthogonal trajectory exists for $a = 1$. Correspondingly, the patterns are labeled by the symbols $P4$, $P2$, R 1, $S2$. In contrast to the situation in Sec. III B, the case $a = 0$ is also of type $P4$ here.

3. The x axis is the longer principal axis of the difference cone (c < 1)

a. Consider first the case $1 < c < \sqrt{2}$. This case differs from the preceding case in two respects. First, for $a = 0$, the long axis of the elliptical cone is now along the x axis. Second, Eq. (3.14b) now yields $\mathrm{sign}(x) > 0$, so that the two straight-line orthogonal trajectories with $k \neq 0$ are inclined to the right. These features are illustrated in Fig. 5, depicting the case $c = 1.25$, which otherwise is entirely analogous to

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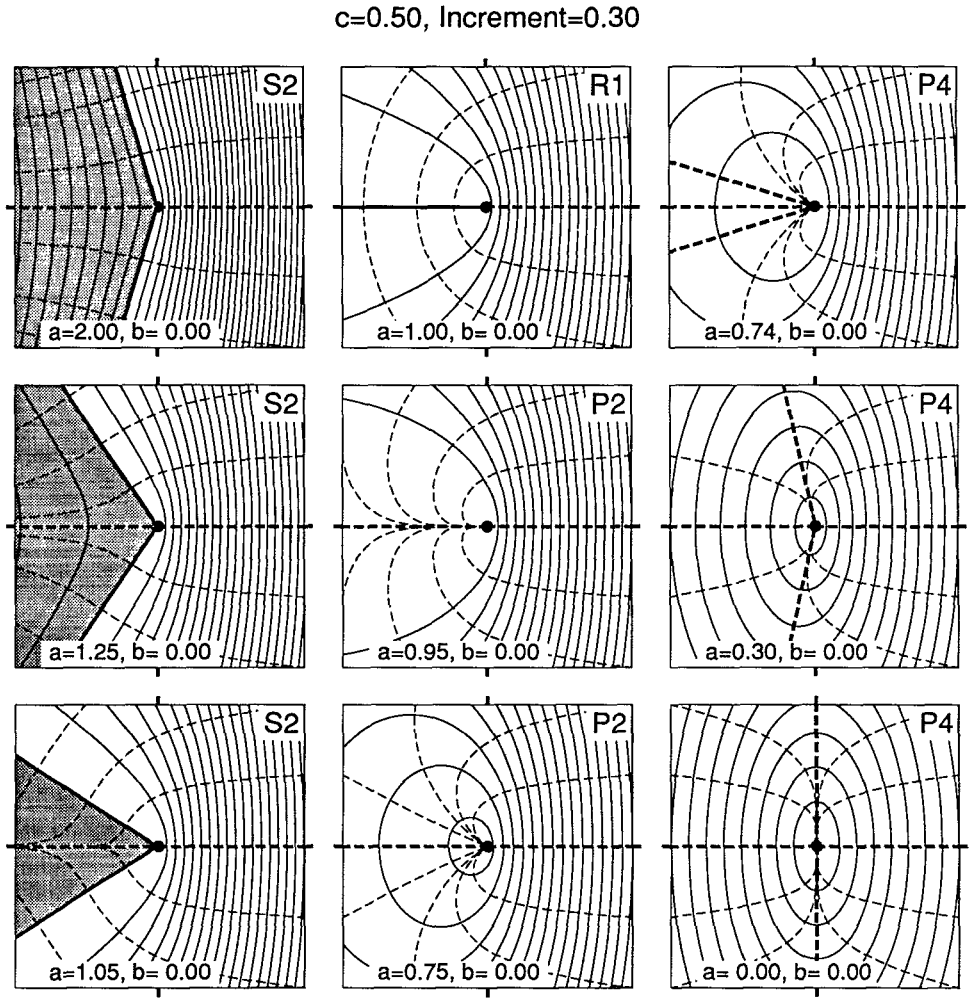

FIG. 4. Potential energy surface patterns near an intersection for the case $c = 0.5$, $b = 0$. For explanations of curves, see the caption of Fig. 3.

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Fig. 4 for $c < 1$. According to Eq. (3.14a), the two straight-line orthogonal trajectories collapse into the positive x axis for $a = |c^2 - 1| = c^2 - 1 = 0.5625$.

b. Next, consider the case $c > \sqrt{2}$. It is illustrated in Fig. 6 for $c = 1.7$. This case has an interesting further feature in addition to those shown on Fig. 5. It arises from the fact that, according to Eq. (3.14a), the two straight-line orthogonal trajectories with $k \neq 0$ will collapse into the positive x axis only for $a = |c^2 - 1| = (c^2 - 1) > 2 - 1 = 1$; in the present example, for $a = 1.89$. Thus, there exist now patterns where $1 < a < |c^2 - 1|$, i.e., where the surface has a region with $f > 0$ and *also* two straight-line orthogonal trajectories with $k \neq 0$. These surfaces are illustrated on the three panels labeled $S4$ on Fig. 6. For $a = 1$, the negative x axis is no longer an orthogonal trajectory and it is therefore labeled $R3$.

The surface patterns of type $S4$ thus exhibit the following new pattern: one downhill steepest-descent line enters the intersection point from the left along the negative x axis, but *infinitely many* steepest-descent lines leave the intersection point towards the right, initially parallel to the positive x axis, but quickly fanning out between the two bold-faced straight-line orthogonal trajectories with $k \neq 0$ as indicated by the lightly shaded area in Fig. 6. This pattern thus presents a situation where a steepest-descent line that could model a reaction path can fan out into an infinite number of steepest-descent lines upon passing through an intersection point. This possibility further emphasizes the bifurcating character of the intersection region should the intersection point lie on a reaction path.

D. The general case ($c \neq 1$ and $a,b \neq 0$)

In general, the surface pattern of $f(x,y)$ has no symmetry and a more complicated structure than in the two special cases considered in Secs. III B and III C. Nonetheless, it remains true that the chief criteria characterizing the various topographies are (i) whether there exist straight-line contours through the origin on which $f(x,y)$ vanishes and (ii) how many straight-line orthogonal trajectories through the origin there are. We shall examine the analytic conditions pertaining to these two criteria.

1. Straight-line contours $f = 0$ through the origin

From Eq. (3.5), it is apparent that $f(x,y)$ always vanishes at the origin and that there is always a region where $f(x,y)$ is negative. Two fundamentally different general patterns are therefore possible, namely, *regions where $f(x,y)$ is positive do exist or such regions do not exist.* The former case can be identified by observing that there must exist *contour curves* which separate the domain where $f > 0$ from the do-

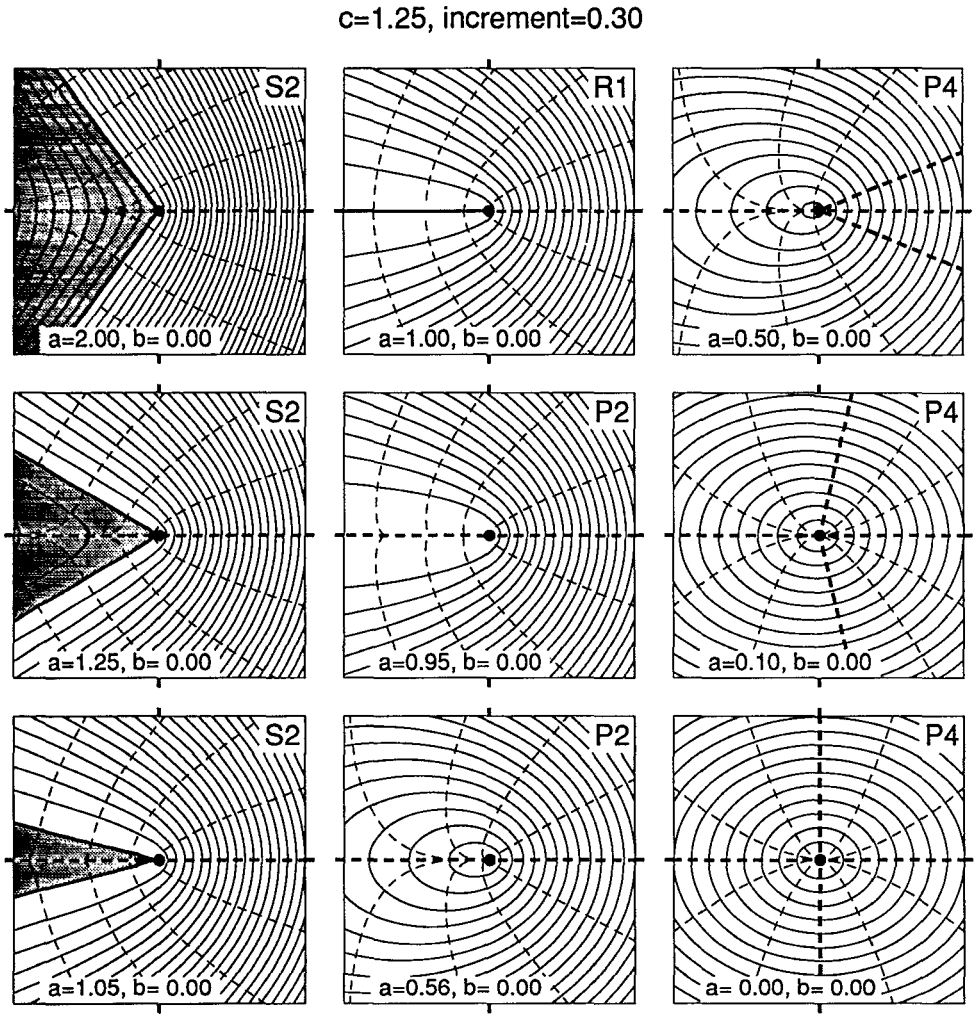

FIG. 5. Potential energy surface patterns near an intersection for the case $c = 1.25$, $b = 0$. For explanations of curves, see the caption of Fig. 3.

main where $f<0$. Along such curves, $f(x,y)$ vanishes, which implies that

$$-ax - by = (x^2 + c^2y^2)^{1/2}. \tag{3.15}$$

Since this equation is of the form $F(y/x) = 0$, it determines one or several straight lines $y/x = m =$ constant. The function f vanishes therefore *always* on straight lines passing through the origin. Their slopes m are obtained as solutions of the quadratic equation

$$(c^2 - b^2)m^2 - 2abm - (a^2 - 1) = 0 \tag{3.16}$$

where

$$m = (ab \pm \sqrt{D})/(c^2 - b^2), \quad D = b^2 + c^2(a^2 - 1). \tag{3.17}$$

The coefficients a, b, and c and the discriminant D are thus related by

$$a^2 + b^2/c^2 = 1 + D/c^2 \tag{3.18}$$

and three cases are possible.

a. $D>0$, $a^2 + b^2/c^2 > 1$. There exist two different solutions m_1 and m_2 and, hence, two straight lines. However, because of Eq. (3.15), the inequality

$$ax + by = (a + bm)x < 0 \tag{3.19}$$

must also be satisfied. It implies that for given values of a, b, and m, the values which x may assume are limited to having the opposite sign as $(a + bm)$. Hence m_1 and m_2 generate only half-lines, both of which end at the origin. Together they form the contour $f = 0$ which separates the domain where $f>0$ from that where $f<0$. As before, we shall again use the symbol S to label such patterns.

b. $D = 0$, $a^2 + b^2/c^2 = 1$. The two lines of the previous case coalesce into one line with $m = ab/(c^2 - b^2) = b/ac^2$. Condition (3.19) therefore yields the result $x/a < 0$, i.e., that x must have the opposite sign as a. Since a is assumed to be positive, we thus always have one half-line with $x < 0$ which ends at the origin. The region where $f>0$ has again been squeezed out. The function $f(x,y)$ assumes its maximum value $f = 0$ on the mentioned half-line. As before, we shall use the label R for this pattern.

c. $D<0$, $a^2 + b^2/c^2 < 1$. In this case, $f(x,y)$ vanishes only at the origin and this is the point where $f(x,y)$ reaches its maximum. $f(x,y)$ is negative everywhere else. As before, we shall label this pattern by the symbol P.

2. Straight-line orthogonal trajectories through the origin

From Eq. (3.5) it follows that the orthogonal trajectories of $f(x,y)$ are functions $y(x)$ which are defined by the differential equation

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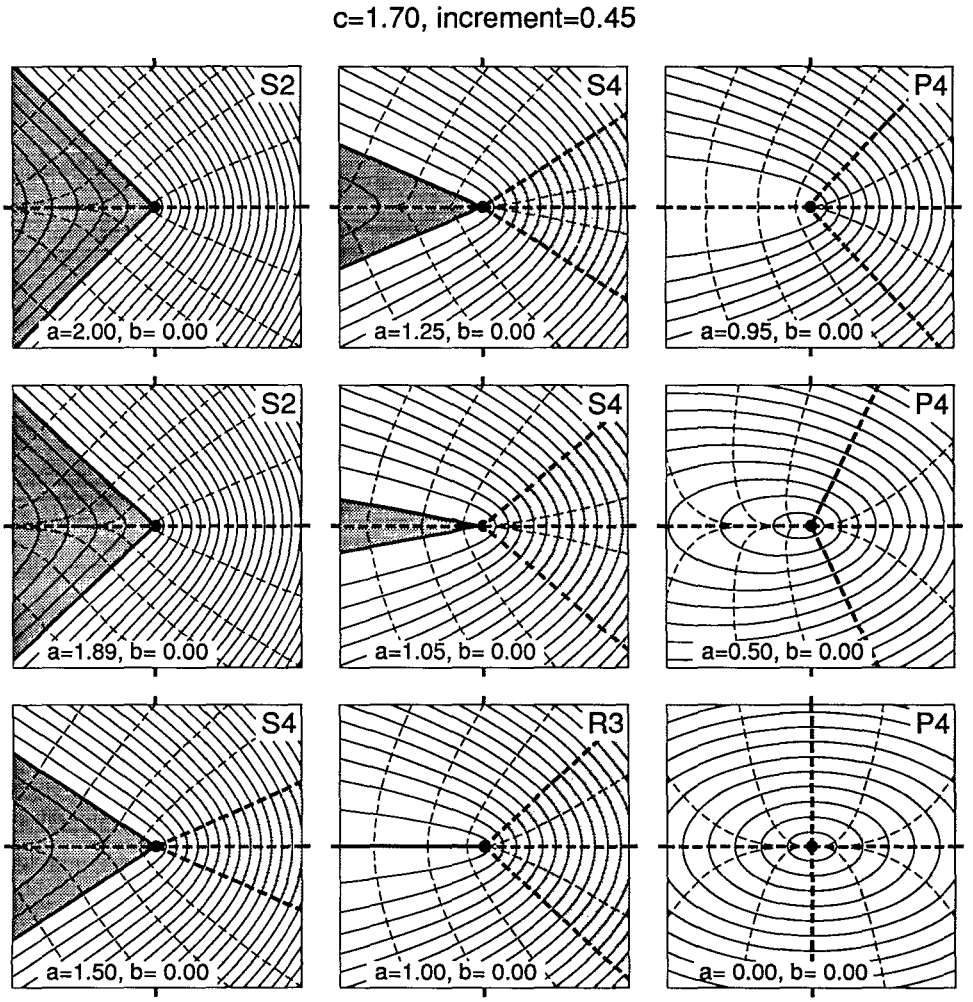

FIG. 6. Potential energy surface patterns near an intersection for the case $c = 1.70$, $b = 0$. For explanations of curves, see the caption of Fig. 3, except that the increment between contours is 0.45 in this figure.

$$dy/dx = f_y/f_x = (br + c^2y)/(ar + x) \tag{3.20a}$$

with

$$r = (x^2 + c^2y^2)^{1/2}. \tag{3.20b}$$

This differential equation is of the homogeneous form

$$dy/dx = f(t), \quad t = y/x, \tag{3.21}$$

$$f(t) = [b(1 + c^2t^2)^{1/2} + c^2t]/[a(1 + c^2t^2)^{1/2} + 1], \tag{3.22}$$

which can be reduced to a quadrature in t. The latter yields the parametric representation

$$x = K \exp\{\int dt\, [f(t) - t]^{-1}\}, \quad y = xt, \tag{3.23}$$

where different values of the integration constant K determine different orthogonal trajectories.

In general, the orthogonal trajectories are curvilinear and, with appropriate changes of the parameter values a, b, and c, their curvatures can invert. Whenever this happens, a separating orthogonal trajectory is found which is a straight line passing through the origin: $y = kx$. The condition for this to happen is obtained by inserting $y = kx$ into Eq. (3.20). This yields

$$[b(x^2 + c^2k^2x^2)^{1/2} + c^2kx]/[a(x^2 + c^2k^2x^2)^{1/2} + x] = k = \text{constant}$$

from which follows

$$ak - b = \sigma g(k), \tag{3.24a}$$

where

$$\sigma = \text{sign}[x(c^2 - 1)], \tag{3.24b}$$

$$g(k) = |c^2 - 1|k/(1 + k^2c^2)^{1/2}, \tag{3.25}$$

i.e.,

$$ak - b = g(k), \quad \text{when sign}(x) = \text{sign}(c^2 - 1), \tag{3.26}$$

$$-ak + b = g(k), \quad \text{when sign}(x) = -\text{sign}(c^2 - 1). \tag{3.27}$$

Equations (3.26) and (3.27) show that the straight lines $y = kx$ end at the origin, i.e., they are again half-lines. Equations (3.24)–(3.27) are equations of the fourth degree for the slope k. The nature of their solutions can be recognized from the graphic representation in Fig. 7 which exhibits a plot of the function $g(k)$ of Eq. (3.25) vs k for some arbitrary value of $c(\neq 1)$ as well as graphs of three straight lines $(ak - b)$, denoted by A, B, and C, and three straight lines $(-ak + b)$, denoted by A', B', and C'. The abscissas of the intersections of any one of these straight lines with the function $g(k)$ are the solutions of Eq. (3.24) for the pertinent values of a, b, and c. The lines A', B', and C' yield the solu-

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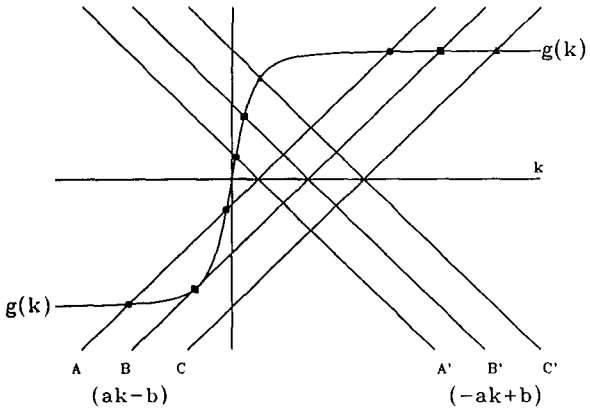

FIG. 7. A plot of the function $g(k)$ defined in Eq. (3.25), of three lines $(ak-b)$ and three lines $(-ak+b)$, where $a,b>0$. The intersections yield solutions of Eq. (3.24).

tions of Eq. (3.27) and it is apparent that there always exists one such solution. The lines A, B, and C yield the solutions of Eq. (3.26) and there can be manifestly one, two, or three such solutions. Since we have $b \neq 0$, the following conclusions can be drawn.

First, there always exist two solutions with $k>0$. In most cases, they correspond to two straight-line orthogonal trajectories with positive slopes, one being a half-line with $\mathrm{sign}(x)=\mathrm{sign}(c^2-1)$, the other being a half-line with $\mathrm{sign}(x)=-\mathrm{sign}(c^2-1)$. The only exception is the case when a, b, and c satisfy the condition $a^2+b^2/c^2=1$, which leads to the pattern R, discussed after Eq. (3.19), when the two straight-line *contours* $f=0$ collapse into one such contour with $x<0$ and the slope $m=b/ac^2$. It turns out that this limiting straight-line contour in fact satisfies Eq. (3.24). Indeed, replacing k by $m=b/ac^2$ and $\mathrm{sign}(x)$ by (-1) in Eq. (3.24), one finds

$$a(b/ac^2)-b=\mathrm{sign}(1-c^2)|1-c^2|(b/ac^2)/(1+c^2b^2/a^2c^4)^{1/2},$$

which simplifies to

$$1=1/(a^2+b^2/c^2)^{1/2},$$

an identity that is in fact satisfied. This result implies that, in this case, the solution $k>0$ of Eq. (3.24) with $\sigma=\mathrm{sign}(1-c^2)$ is a *contour* $f=$ constant and hence *not a steepest-descent line*. It is also readily verified that f_y/f_x is no longer given by Eq. (3.20) in this case. Thus, the pattern R always has one straight-line orthogonal trajectory less than the corresponding patterns P and S. The same result had been found for the specific cases considered in Secs. III A and III C.

Second, there can exist two, one, or no solutions with $k<0$. They yield straight-line orthogonal trajectories with negative slopes, all of which are half-lines with $\mathrm{sign}(x)=\mathrm{sign}(c^2-1)$. The condition for having two, one, or no such trajectories is obtained as follows. According to Fig. 7, the limiting case of *one* solution with $k<0$ corresponds to the line B which is tangent to the curve $g(k)$. Equating the slopes of $(ak-b)$ and $g(k)$ yields

$$a=|c^2-1|/(1+k^2c^2)^{3/2} \tag{3.28}$$

and this equation, together with Eq. (3.26), embodies the requirements for tangency. It is possible to eliminate the variable k between Eqs. (3.26) and (3.28) and, thereby, one obtains the condition

$$(a^2)^{1/3}+(b^2c^2)^{1/3}=[(1-c^2)^2]^{1/3} \tag{3.29}$$

among a, b, and c. By reference to the intercept b in Fig. 7, it is readily seen that

no solution $k<0$ exists when

$$|a/(c^2-1)^{2/3}+|bc/(c^2-1)|^{2/3}>1, \tag{3.30a}$$

one solution $k<0$ exists when

$$|a/(c^2-1)|^{2/3}+|bc/(c^2-1)|^{2/3}=1, \tag{3.30b}$$

two solutions $k<0$ exist where

$$|a/(c^2-1)|^{2/3}+|bc/(c^2-1)|^{2/3}<1. \tag{3.30c}$$

3. Classification of patterns

The distinctions made in the preceding subsections are combined graphically in Fig. 8, which displays plots of

$$a^2+b^2/c^2=1 \quad \text{(dashed curves)} \tag{3.31}$$

as well as plots of

$$|a/(c^2-1)|^{2/3}+|bc/(c^2-1)|^{2/3}=1 \quad \text{(solid curves)} \tag{3.32}$$

in the a–b plane for various values of the parameter c. Four panels are shown corresponding to the four cases $0<c<1/\sqrt{2}$, $1/\sqrt{2}<c<1$, $1<c<\sqrt{2}$, $\sqrt{2}<c$. Depending on the value of c, the two curves divide the first quadrant into five or nine regions which are labeled by the symbols introduced earlier to denote the various surface patterns that can

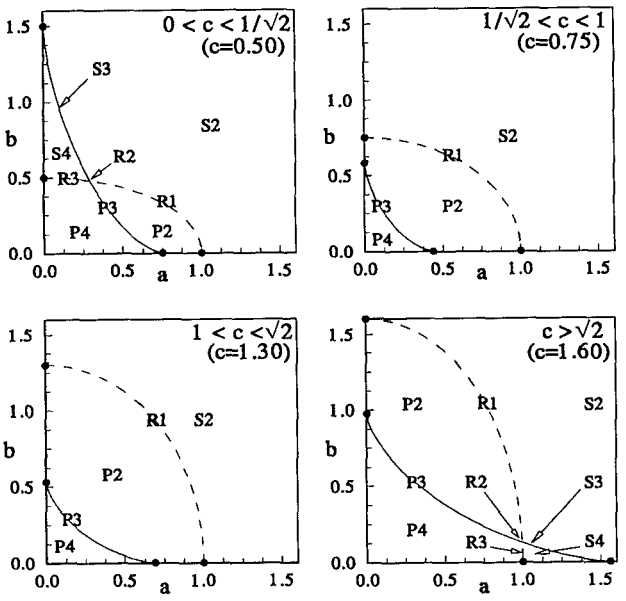

FIG. 8. Graphs in the a–b plane for the classification of potential energy surface patterns near an intersection. Solid curves = plots of Eq. (3.32); dashed curves = plots of Eq. (3.31). The symbols in various regions of the quadrant are defined in Table I. Symbols on curves refer to those curves.

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TABLE I. Patterns for the function $f(x,y)$ of Eq. (3.5) when $c \neq 1$, $a \neq 0$, and $b \neq 0$.

	a^2+b^2/c^2	$\lvert bc/(c^2-1)\rvert^{2/3}$ + $\lvert a/(c^2-1)\rvert^{2/3}$	Dimension of the region where $f>0$	Number of straight-line orthogonal trajectories		
				$k>0$, $x<0$	$k>0$, $x>0$	$k<0$, $x(c^2-1)>0$
$P2$	<1	>1	0	1	1	0
$P3$	<1	$=1$	0	1	1	1
$P4$	<1	<1	0	1	1	2
$R1$	$=1$	>1	1	0	1	0
$R2$	$=1$	$=1$	1	0	1	1
$R3$	$=1$	<1	1	0	1	2
$S2$	>1	>1	2	1	1	0
$S3$	>1	$=1$	2	1	1	1
$S4$	>1	<1	2	1	1	2

result from the corresponding parameter choices. The characteristics of the nine different types are listed in Table I.

On the panels of Fig. 8, the patterns $P2$, $P4$, $S2$, and $S4$ obtain in two-dimensional regions, the patterns $P3$, $S3$, $R1$, and $R3$ obtain on curves, and the pattern $R2$ obtains at points. It is apparent from Eq. (3.5) that the function $f(x,y)$ for $(a=u,\ b=v,\ c=w)$ can be obtained from the function for $(a=v/w,\ b=u/w,\ c=1/w)$ by interchanging x and y and multiplying the entire function by w. Thus, for every point with $a/b=\alpha$ on a panel for c, there exists a point with $a/b=\alpha^{-1}$ on the panel for c^{-1} in Fig. 9 such that the corresponding surfaces differ essentially only in the interchange of x and y. Therefore, if we consider $c>1$ as well as $c<1$, then we have to discuss only the patterns for $a>b$. Conversely, if we consider all values a and b of the panel for c, we do not have to discuss the panel for c^{-1}.

Figure 9 depicts nine PES contour plots for $c=2.29$, corresponding to the nine patterns listed on the panel for

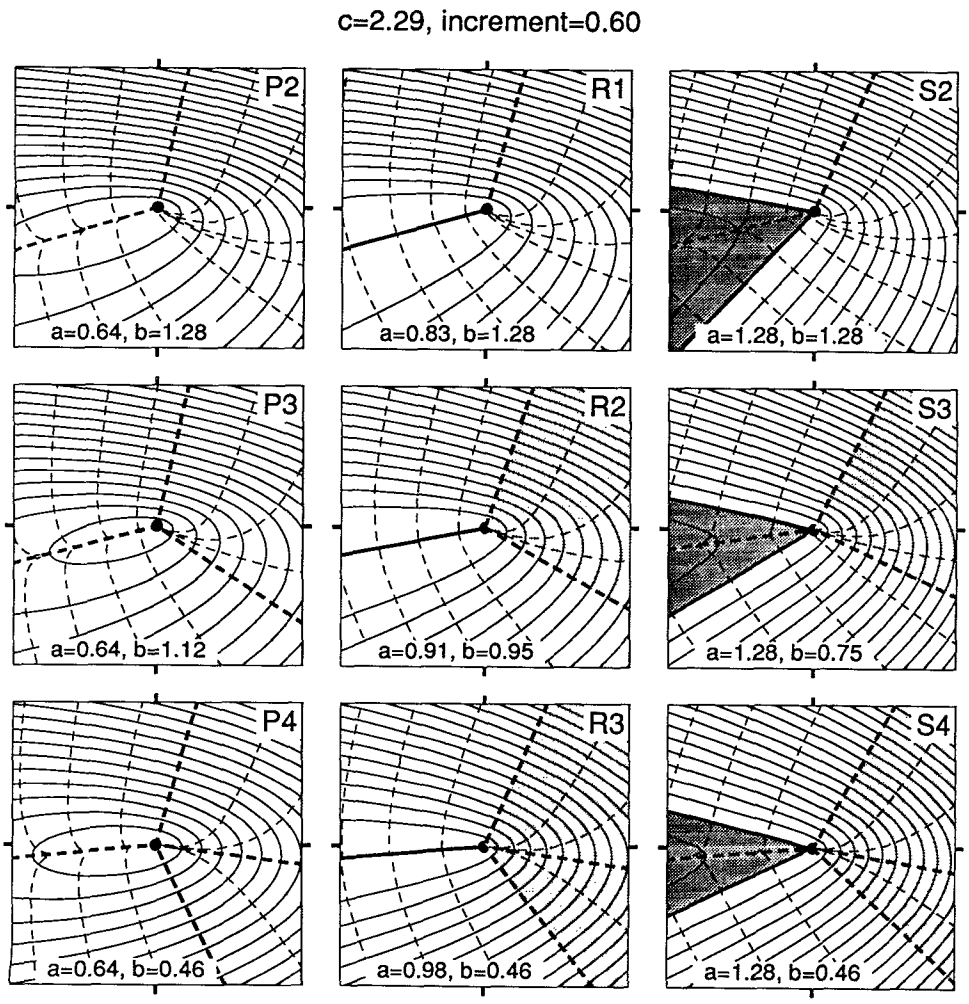

FIG. 9. Potential energy surface patterns near an intersection for the case $c=2.29$, $a \neq 0$, $b \neq 0$. For explanations of curves, see the caption of Fig. 3, except that the increment between contours is 0.6 in this figure.

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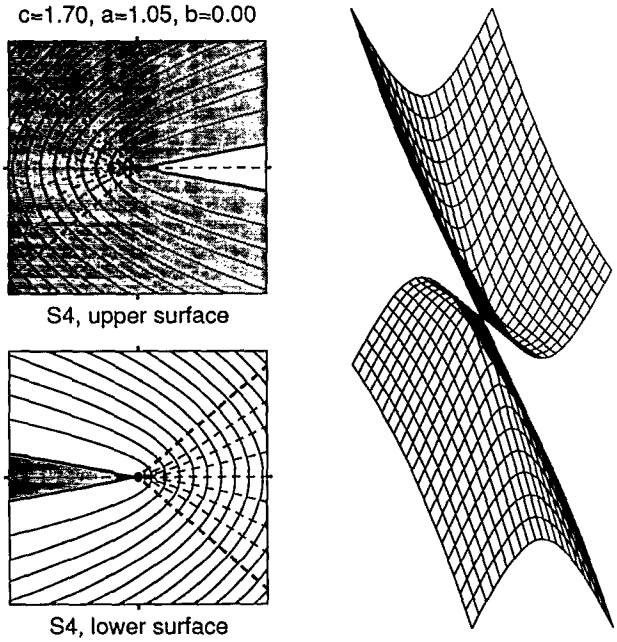

FIG. 10. Contour maps (left) and perspective representation (right) of potential energy surfaces of two states in the vicinity of their intersection (Pattern S4). Left figures: solid lines = contours; increment = 0.55. On both panels the contour values decrease from left to right. The bold-faced solid straight lines are the contours $E = E_{\text{intersection}}$. Shaded areas = regions where $E > E_{\text{intersection}}$. Dashed lines = steepest descent curves passing through the intersection. Only the bold-faced dashed lines and the x axis are straight; all other orthogonal trajectories become tangent to the x axis at the intersection. The x axis goes from left to right on the contour maps and from back left to front right on the perspective figure.

$c > \sqrt{2}$ in Fig. 8. The panels on Fig. 9 are arranged relative to each other in the same order as the regions corresponding to these patterns on the $c > \sqrt{2}$ panel of Fig. 8. As before, solid bold-faced lines are contours $f = 0$ and dashed bold-faced lines are straight-line orthogonal trajectories. The dark shaded areas to the left of the intersection on the S panels are regions where $f > 0$. The light shaded areas to the right of the intersection on the panels $S3$, $S4$, $R2$, and $R3$ are regions where infinitely many orthogonal trajectories leave the origin towards the right and fan out between the bold-faced dashed lines. It is apparent that the bunching of the orthogonal trajectories near the origin differs qualitatively for the three cases characterized by Eqs. (3.30a), (3.30b), and (3.30c).

The patterns which one obtains for $c < 1$ differ from those of Fig. 9 in that the relative positions of the long and the short axes of the difference cone are interchanged with respect to the bold-faced contours and trajectories.

IV. TOPOGRAPHY OF THE UPPER SURFACE

From Eq. (3.5), we know that near an intersection, the surface of the upper state can be obtained from that of the lower state by the relation

$$E_{\text{upper}}(x,y) = -E_{\text{lower}}(-x,-y). \tag{3.33}$$

In the case of S patterns, it is seen that both states have the same overall direction of their slopes and, at the intersection, merely exchange the steepness of their slopes. It is also apparent from Eq. (3.33) that the upper state has a valley floor wherever the lower state has a ridge. Consequently, whereas downhill reaction paths on E_{lower} will tend to veer away from the crossing, such reaction paths on E_{upper} will be funneled towards the crossing. As an example, Fig. 10 displays contours and orthogonal trajectories of both states for the case $S4$ discussed in Sec. III C. The plot of the lower surface is identical with that for $a = 1.05$ on Fig. 6 except that the contour increment has been increased. The plot of the higher surface is obtained from it by Eq. (3.33). On both panels, the shaded area indicates where the energy is higher than the intersection energy, as before. For clarity, only steepest-descent lines passing through the intersection are drawn (dashed lines).

The negative x axis is seen to be the floor of a valley on the upper surface descending to the intersection, whereas the positive x axis is the crest of a ridge on the lower surface descending from the intersection. It is apparent that on the upper surface, there exists a funnel of steepest-descent lines (shown in Fig. 10) that are focusing toward the intersection and emerge on the lower surface where they fan out in a diverging pattern. On each of these steepest-descent lines, the wave function of the one state goes continuously into that of the other state, as discussed in Sec. II C, and for these *diabatic* paths, the intersection is a diabatic valley–ridge inflection point.[7] It is apparent that these conditions strongly favor radiationless transitions from the upper surface to the lower surface. Outside the funnel of steepest-descent lines shown in Fig. 10, the steepest-descent lines of the upper surface do not reach the intersection and stay on the upper surface.

On the other hand, on the lower surface, the negative x axis is a ridge descending towards the intersection and the steepest-descent line on this ridge emerges on the upper surface as the floor of a valley along the positive x axis. All other steepest-descent lines on the lower surface veer, however, away from the intersection and stay on the lower surface.

In the case of P patterns, the lower state has a pointed maximum at the intersection, whereas the upper state has a pointed minimum. Thus *all* steepest-descent lines of the upper surface lead to the intersection and emerge on the lower surface. There will be an even stronger tendency to funnel reaction paths (should there be one) towards the crossing with a resultant radiationless transition to the lower state.

In some of the cases considered, radiationless transitions from E_{lower} to E_{upper}, although less likely, may also occur.

V. CONCLUSIONS

The topography of potential energy surfaces in the immediate vicinity of a crossing with another surface can have two main patterns—a peaked (P) pattern or a sloped (S) pattern. For S patterns, both PES's have downhill slopes and touch each other at the crossing points in the branching space. For P patterns, both PES's are elliptical cones pointing against each other with a common tip. A limiting pattern, called R, intermediate between the P and S patterns, is also possible where there exists a line along which E_1 has a zero slope on one side of the intersection and E_2 has zero

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slope on the other side of the intersection. A considerable variety of patterns can occur for the steepest-descent lines. In some, but not in all cases, infinitely many steepest-descent lines enter or leave the intersection in certain, but not necessarily in all directions.

The overall character of the steepest-descent lines is such that on the lower PES, reaction paths will be steered away from the crossing and tend to bifurcate in its vicinity, whereas on the upper surface, they will tend to be funneled towards the intersection with a strong probability for a dynamic transition from one state to the other because the two states exchange character (continuity of energy slope and of wave function from one state to the other) on *any* straight path through the intersection. It can therefore be inferred that intersections are preferred locations for radiationless transitions from the higher state to the lower state. For S patterns, radiationless transitions from the lower state to the higher state are also conceivable on the downhill slope of both energy surfaces.

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